MODULE 10 / LESSON 01

Preprocessing Is Part of the Model

Raw tables often need preprocessing before a model can use them:

DSACADEMY

CORE CONCEPT

Imputation Learns a Replacement Rule

Imputation replaces declared missing values using a rule. Common simple rules: The replacement value is a learned parameter. Fit it on training data, then apply the same value to validation, test, and production rows.

numerical median;

numerical mean;

most frequent category;

constant marker such as "missing".

RUN IT AND INSPECT THE RESULT

Scaling Changes Units, Not Row Meaning

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
scaler.fit(X_train[["income"]])
```

OUTPUT

The scaler stores training mean and variance for later transforms.

The value is two standard deviations above the training mean.

REASONING SEQUENCE

Encoding Represents Categories Numerically

Frame

Models usually require numerical inputs.

1

Transform

One-hot encoding creates one binary column for each known category.

2

Validate

City values Colombo, Kandy, and Galle become columns such as:

3

Explain

cityColombo, cityKandy, cityGalle

4

Kandy becomes: [0, 1, 0] The three positions correspond to the fitted category order. If Jaffna was unseen, handleunknown="ignore" produces zeros for known category columns.

CORE CONCEPT

ColumnTransformer Applies Rules by Column Type

A ColumnTransformer applies different transformer sequences to selected columns and concatenates their outputs. Example: The transformed feature matrix may contain more columns than the raw table because one category column expands into several indicator columns.

numerical columns -> median imputer -> standard scaler;

categorical columns -> most-frequent imputer -> one-hot encoder.

RUN IT AND INSPECT THE RESULT

Pipeline Gives One Fit and Predict Boundary

```
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression

model = Pipeline([
    ("preprocess", preprocess),
    ("classifier", LogisticRegression(max_iter=1000)),
])
```

OUTPUT

A composite estimator exposing fit, predict, and predict_proba.

One call to fit() learns imputers, scaler, encoder, and classifier. One call to predict() reuses all learned pieces in order.

CORE CONCEPT

Pipelines Prevent Preprocessing Leakage

Unsafe sequence: Validation information influences the scaler. Safe sequence:

fit scaler on the complete dataset;	**transform all rows;**	**split or cross-validate.**
split training and test data;	**put scaler and estimator inside a pipeline;**	**for each validation fold, fit the complete pipeline only on that fold's training rows;**

REASONING SEQUENCE

Cross-Validation Measures Variation Across Splits

Frame

split training data into k folds;

1

Transform

fit on k-1 folds;

2

Validate

validate on the remaining fold;

3

Explain

repeat until every fold has served as validation;

4

Five validation scores: [0.72, 0.75, 0.68, 0.78, 0.71] Mean: 3.64 / 5 = 0.728 Range 0.68 to 0.78 shows split-to-split variation.

RUN IT AND INSPECT THE RESULT

Model Comparison Must Use the Same Evidence

```
from sklearn.model_selection import GridSearchCV

search = GridSearchCV(
    model, {"classifier__C": [0.1, 1, 10]},
    cv=5, scoring="roc_auc"
)
```

OUTPUT

Each C value is evaluated through the complete pipeline across five folds.

The tree's slightly higher mean comes with greater variation. The correct choice requires more than the largest mean.

QUALITY GATE

Reproducibility Requires More Than a Random Seed

- [x] code revision;
- [x] immutable data identity or checksum;
- [x] feature and target schema;
- [x] split method and random seed;

QUALITY GATE

Persist the Whole Artifact and Load It Carefully

- [x] complete preprocessing and estimator pipeline;
- [x] input schema;
- [x] label meanings and threshold;
- [x] dependency versions;

GUIDED LAB

Guided Lab: Build a Leakage-Safe Pipeline

01

numerical and categorical column lists

02

missing-value policies

03

numerical scaling

04

categorical encoding with unknown handling